ELEC60021
ELEC97054

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2022

MATHEMATICS FOR SIGNALS AND SYSTEMS

Wednesday, 7 December 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions. All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : D. Angeli

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Special Information for the Invigilators: none

Information for Candidates:

For questions involving matrix manipulations or calculations, please provide answers showing intermediate calculations steps.

A Hermitian matrix $\mathbf{A}$ is positive definite if and only if:

- All the eigenvalues satisfy $\lambda_i > 0$
- All the pivots (without row exchange) satisfy $d_i > 0$

The Questions

1. (a) Consider the system $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{A} \in \mathbb{R}^{3\times 3}$ and $\mathbf{b} = [9, -4.5, -4.5]^T$. The complete solution to this system is

$$\mathbf{x} = \begin{bmatrix} 1 \\ -1 \\ 5 \end{bmatrix} + a \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$$

for arbitrary a and b.

i. Find the rank of $\mathbf{A}$. [2]

ii. Find $\mathbf{A}$ [2]

iii. Find the minimum norm solution. [2]

(a) You want to approximate a function $y(t)$ with a polynomial $p(t)$. You observe $y(t)$ at instants $t_1 = 0, t_2 = 1, t_3 = 2, t_4 = 4$ and $t_5 = 5$ and we have that $y_1 = y(t_1) = -1$, $y_2 = y(t_2) = 0.5$, $y_3 = y(t_3) = 0.5$, $y_4 = y(t_4) = 2$ and $y_5 = y(t_5) = 1.5$.

i. Assume you want to find a polynomial of degree two that satisfies $p(t_1) = y_1$, $p(t_4) = y_4$ and $p(t_5) = y_5$. Can you claim that this polynomial exists and is unique? Briefly justify your answer. [4]

ii. Find a polynomial $p_1(t) = a_1 t + b_1$, that satisfies $p_1(t_1) = y_1$ and $p_1(t_4) = y_4$. [2]

iii. Find now a polynomial $p_2(t) = a_2 t + b_2$ that minimises

$$e_2 = \sum_{i=1}^{5} |y_i - a_2 t_i - b_2|^2.$$

[4]

iv. Can you claim that the solution you found is unique? [2]

v. Verify that $e_2 < \sum_{i=1}^{5} |y_i - a_1 t_i - b_1|^2$ [2]

2. (a) Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 5 & 1 \\ 2 & 1 & 2 \end{bmatrix}.$$

You are told that the largest singular value of $\mathbf{A}$ is $\sigma_1 = 6$, the singular vector $\mathbf{u}_1 = [1/\sqrt{6}, 2/\sqrt{6}, 1/\sqrt{6}]^T$ and $\mathbf{v}_2 = [1/\sqrt{3}, -1/\sqrt{3}, 1/\sqrt{3}]^T$.

i. Find all the singular values of $\mathbf{A}$ [2]

ii. Find the SVD of $\mathbf{A}$ [4]

iii. Find the least-square minimum norm solution to $\mathbf{Ax} = \mathbf{b}$ with

$$\mathbf{b} = [9, 9, -9]^T.$$

[4]

(b) Consider the system of linear equations $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{A}$ given by:

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 0 & c \end{bmatrix}.$$

i. Assume $c = 0.5$, can you guarantee that the system has always at least one solution? Justify your answer. [2]

ii. Assume now that $c = 1$ and that $\mathbf{b} = [-1, 2]^T$, find the minimum norm solution to $\mathbf{Ax} = \mathbf{b}$. [2]

(c) Try to solve the following constrained minimisation. Given the vector $\mathbf{y} = [1, 1, -1]^T$, find the three dimensional vector $\mathbf{x} = [x_1, x_2, x_3]^T$ that is closest in the least-squares sense to $\mathbf{y}$ among all three-dimensional vectors that have average value $\mu = 1$. That is find the $\mathbf{x}$ that minimises $\|\mathbf{y} - \mathbf{x}\|^2$ subject to $(x_1/3 + x_2/3 + x_3/3) = 1$. [4]

3. (a) Consider the matrix

$$\mathbf{Q} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Is $\mathbf{Q}$ positive definite and why? [2]

(b) Let $\boldsymbol{Q}$ be an $n \times n$ positive definite matrix. The directions $\boldsymbol{d}_0, \boldsymbol{d}_1, ..., \boldsymbol{d}_k \in \mathbb{R}^n$ are called $\mathbf{Q}$-conjugate if for all i, $\boldsymbol{d}_i^T \boldsymbol{Q} \boldsymbol{d}_i \neq 0$, and for all $i \neq j$, $\boldsymbol{d}_i^T \boldsymbol{Q} \boldsymbol{d}_j = 0$. If $\boldsymbol{d}_0, \boldsymbol{d}_1, ..., \boldsymbol{d}_k (k \leq n-1)$ are $\boldsymbol{Q}$-conjugate, show that they are linearly independent. (hint: you can use proof by contradiction.) [5]

(c) Find a set of $\boldsymbol{Q}$-conjugate directions $\boldsymbol{d}_0, \boldsymbol{d}_1, \boldsymbol{d}_2 \in \mathbb{R}^3$ for

$$\boldsymbol{Q} = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

[Hint: you can set $\mathbf{d}_0 = [1, 0, 0]^T$] [3]

(d) We can use the conjugate directions to solve $\boldsymbol{Q}\boldsymbol{x} = \boldsymbol{b}$. We have shown $\boldsymbol{d}_0, ..., \boldsymbol{d}_{n-1}$ form a basis for $\mathbb{R}^n$, therefore, we may express $\boldsymbol{x}$ as $\boldsymbol{x} = \sum_{i=0}^{n-1} \beta_i \boldsymbol{d}_i$, and consequently $\boldsymbol{Q}\boldsymbol{x} = \sum_{i=0}^{n-1} \beta_i \boldsymbol{Q}\boldsymbol{d}_i$.

i. Write the expression for β_i in terms of $\boldsymbol{b}, \boldsymbol{Q}$ and $\boldsymbol{d}_i$. [3]

ii. Given $\mathbf{Q}$ of part (c) and assuming $\mathbf{b} = [6, 0, 2]^T$, find $\mathbf{x}$ using the conjugate directions. [3]

Question 3 continues on the next page

(e) Consider the unweighted and undirected graph shown in Fig. 3.

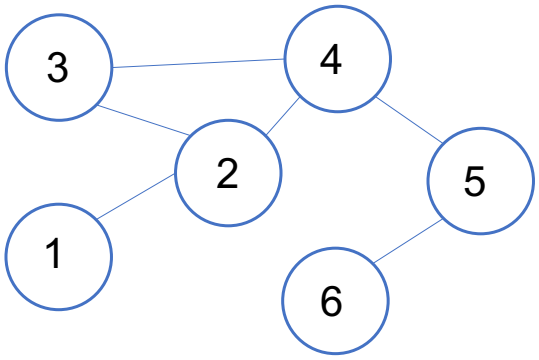


Figure 3: Graph with six nodes

i. Compute the adjacency matrix [2]

ii. Compute two iterations of the power method to estimate the most important node in the graph. [Hint: use the first iteration to estimate the degree of the nodes.] [2]